Descriptive Analytics Report: Banking Sales & Transactions at Scale

Tracking Product Performance, Transaction Velocity, Customer Satisfaction, Regional Demand & Branch Status

Note on data: This report uses a supplied Banking Transaction Dataset workbook. The workbook contains ***300*** category-day observations across 10 product categories, 8 regions, 10 service channels, and 60 tracked branches. The **30-day** transaction data records 1,538,130 transactions and $29,780,780.91 in fee & interest revenue.

TABLE OF CONTENTS

- Objective
- How This Would Work at Real Scale
- Daily Transaction Trend
- Product Category Performance
- Product Trend Velocity
- Customer Satisfaction (CSAT) Distribution
- Regional Demand
- Transaction Value Band & Order Share
- Channel Trends
- Branch Status
- Summary of Key Metrics
- Next Steps to Move From Descriptive → Predictive

1. Objective

Build a descriptive analytics layer over daily banking transactions to answer the recurring business questions that a retail banking, product, or sales-performance team asks every day:

1. Which products and account types are trending or best-selling right now?

2. How are total transactions trending day over day, and what weekly or pay-cycle rhythm sits underneath thattrend?

3. Which product categories are gaining or losing momentum over the course of the month — i.e., what is risingand what is fading?

4. How are customers rating their experience, and where is satisfaction slipping in a way that signals a service, fee,or process problem?

5. Where is banking demand concentrated geographically, and does that concentration match where branchcapacity actually sits?

6. Which branches are at risk of service disruption, and are those the same branches serving high-demandregions?

7. How does transaction value relate to order share — are high-value transactions a meaningful share of volume,or concentrated in a few segments?

8. Which service channels are trending upward or downward, independent of any single product — the signal adigital/channel strategy team needs before the next planning cycle?

2. How This Would Work at Real Scale

Processing daily transactions across a large retail banking portfolio is a data-pipeline problem before it's an analytics problem — but at the scale most retail banks operate at, it does not require streaming infrastructure. A conventional, batch-oriented setup is normal and sufficient

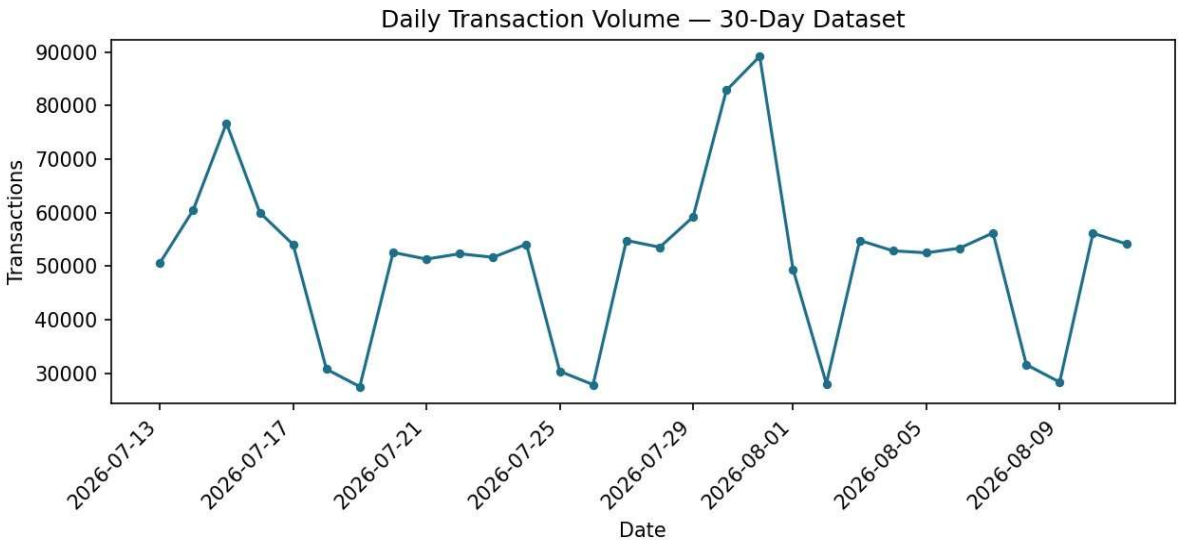
layer	purpose	tools
Data capture	Record each transaction, application, and service event	Point-of-sale system or e-commerce platform export
storage	Store raw transaction records and product/account data	Spreadsheet workbook or shared data file (Excel / Google Sheets)

Cleaning & preparation	**Standardize data, handle missing values, reversals, chargebacks, and duplicates**	**Python (pandas) or Excel formulas run on a schedule**
aggregation	Create daily, category-wise, region-wise, and branch-wise summary tables	Excel pivot tables for smaller catalogs, scheduled scripts for larger ones
scheduling	Automate data cleaning and aggregation tasks	Claude AI
Reporting/Bi	Create dashboards and reports for analysis	Word and Jupyter Notebook

The report below reflects the output of that pipeline — daily aggregated tables — not raw transaction-level data, since no dashboard renders millions of individual line items usefully. Everything downstream (the charts and tables described in the sections that follow) is built on top of these daily summaries.

3. Daily Transaction Trend

The Daily Transaction Trend graph shows total transactions for each of the 30 dates in the dataset. The dataset records **1,538,130 transactions** and **$29,780,780.91 in fee & interest revenue** across the reporting window. The graph shows noticeable variation in daily volume throughout the 30-day period, driven largely by weekday/weekend patterns and pay-cycle dates. The highest daily volume was 89,207 transactions, while the lowest was 27,536 transactions. The 61,671-transaction difference indicates meaningful variation in daily demand.



KPI Definitions Used

- Total Transactions: The total number of transactions recorded during the 30-day reporting period.
- Total Revenue: The total fee and interest revenue generated during the reporting period.
- Highest Daily Volume: The maximum number of transactions on a single day during the reporting period.
- Lowest Daily Volume: The minimum number of transactions on a single day during the reporting period.
- Peak-to-Low Difference: The difference between the highest and lowest daily transaction volume, indicatingvariation in daily demand.
- Daily Transaction Trend: The day-by-day movement of transactions over the 30-day period, helping identifyhigh-demand and low-demand dates.

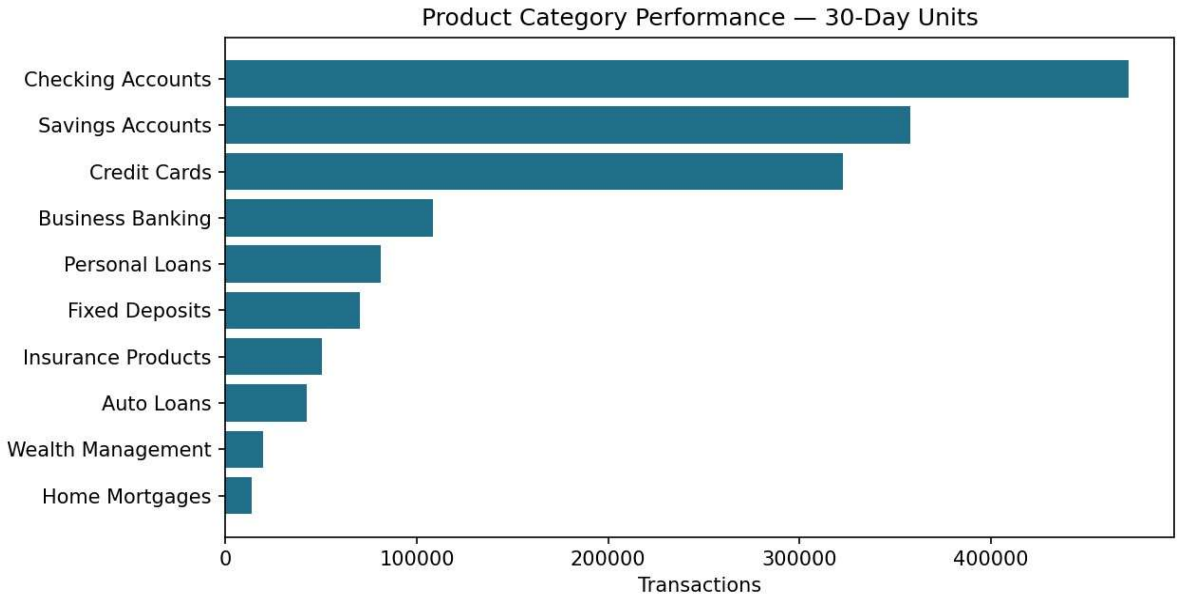
Key numerical summary

Measure	Value
Reporting period	30 days
Total transactions	1,538,130
Total revenue	$29,780,780.91
Highest daily transactions	89,207 (31 Jul 2026)
Lowest daily transactions	27,536 (19 Jul 2026)
Peak-to-low difference	61,671 transactions

Detailed interpretation: The highest daily transaction volume is 89,207 on 31 Jul 2026, while the lowest is 27,536 on 19 Jul 2026. The difference of 61,671 transactions demonstrates meaningful day-to-day variation, consistent with weekday/weekend banking patterns and month-end or payday cycles. Higher points represent stronger observed demand and lower points represent weaker observed demand. This graph helps identify peak dates, slower periods, and unusual movements that may need investigation. It does not by itself prove that a particular promotion, rate change, or marketing campaign caused a spike.

4. Product Category Performance

The Category Performance graph compares transaction volume across the 10 banking product categories. **Checking Accounts** recorded the highest volume with 472,124 transactions, while **Home Mortgages** recorded the lowest with 13,497 transactions. The difference of 458,627 transactions shows a significant variation in category-level demand — expected, since everyday transaction accounts naturally see far higher volume than large, infrequent products like mortgages.



Key numerical summary

Category	30-day units	30-day revenue
Checking Accounts	472,124	$1,125,991.95
Home Mortgages	13,497	$2,819,129.16
All categories	1,538,130	$29,780,780.91

KPI DEFINE USED :

- Category Transaction Volume: Total number of transactions recorded for each banking product category.
- Highest-Volume Category: Product category with the maximum number of transactions.

- Lowest-Volume Category: Product category with the minimum number of transactions

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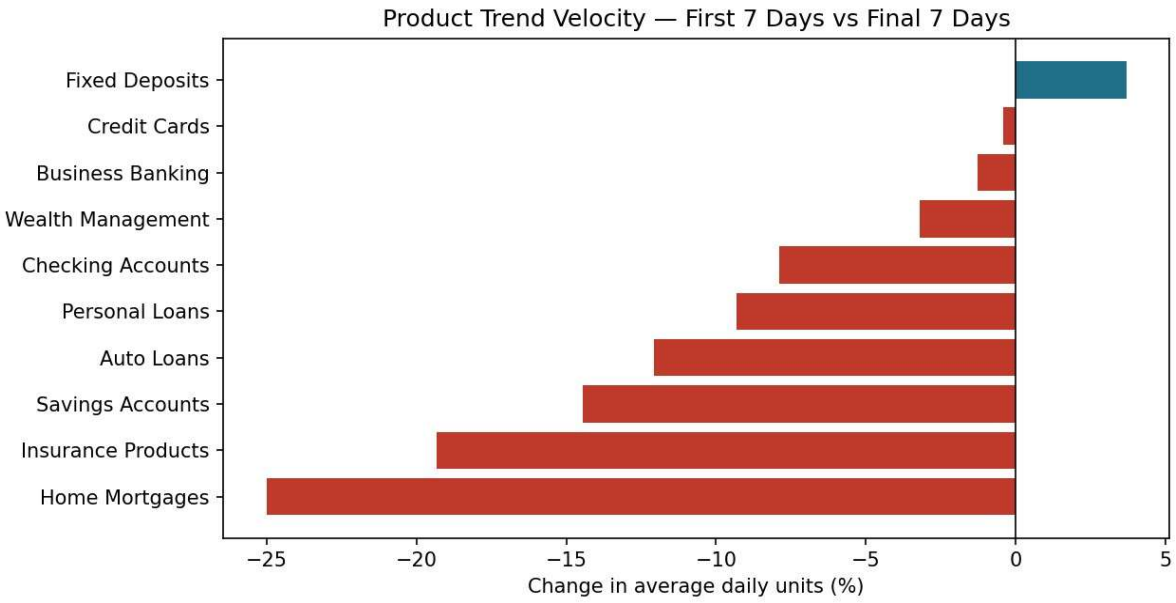
- Category Volume Difference: Difference between the highest- and lowest-volume product categories.

- The report records Checking Accounts as the highest-volume category and Home Mortgages as the lowest.

Detailed interpretation: Checking Accounts is the highest-volume category with 472,124 transactions, while Home Mortgages has the lowest with 13,497. The difference is 458,627 transactions. The bars show where transaction demand is concentrated across product categories. A larger bar means that category contributed more transactions during the 30-day period. Note that revenue does not follow the same ranking as volume — Home Mortgages generated $2,819,129.16 in revenue from far fewer transactions, reflecting a much higher revenue-per-transaction profile typical of large lending products. This supports category comparison and descriptive product-mix discussions, but it does not explain why one category performs better.

5. Product Trend Velocity

Trend velocity measures change over time rather than simply ranking categories by total volume. For this dataset, it compares average daily units during the first seven days with the final seven days. **Fixed Deposits** showed the strongest positive movement, increasing by 3.7%, from an average of 2183.3 units/day in the first seven days to 2263.9 units/day in the final seven days. **Home Mortgages** showed a decline of 25.0%, from 498.7 units/day to 374.0 units/day. This indicates stronger recent demand for Fixed Deposits and a pullback in Home Mortgages during the observed period.



Key numerical summary

Category	First 7-day avg.	Last 7-day avg.	Change
Fixed Deposits	2183.3	2263.9	+3.7%
Home Mortgages	498.7	374.0	-25.0%

KPI DEFINE USED:

First 7-Day Average: Average daily units recorded during the first seven days.

Last 7-Day Average: Average daily units recorded during the final seven days.

Trend Change (%): Percentage change between the first 7-day average and final 7-day average.

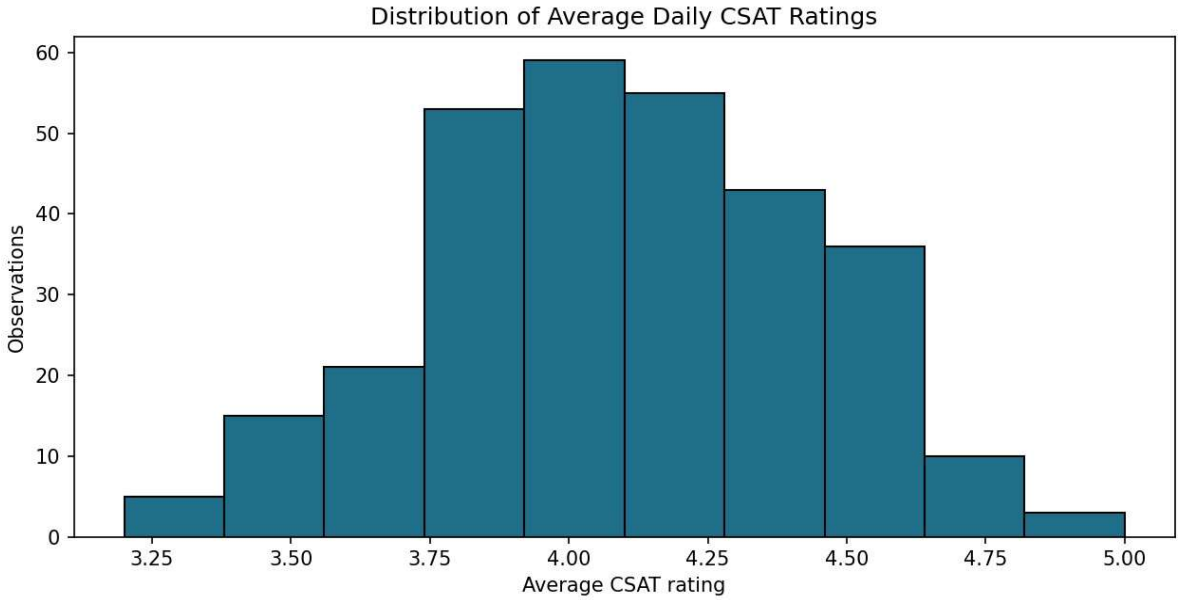
Positive Change: Indicates an increase in observed demand.

Negative Change: Indicates a decrease in observed demand.

Detailed interpretation: Fixed Deposits has the strongest positive movement at +3.7%, increasing from 2183.3 units/day to 2263.9 units/day. Home Mortgages has the largest decline at -25.0%, moving from 498.7 units/day to 374.0 units/day. Positive change indicates stronger recent average demand, while negative change indicates weakening recent average demand. These are descriptive changes within the observed period, not forecasts.

6. Customer Satisfaction (CSAT) Distribution

The Rating Distribution graph shows how reported average daily CSAT ratings are distributed. The overall mean rating is **4.1/5**. Observed ratings range from 3.2 to 5.0 across 300 rating observations. The distribution helps understand overall customer satisfaction patterns and whether ratings are concentrated around higher or lower values.



Key numerical summary

Rating metric	Value
Average rating	4.1 / 5
Minimum observed rating	3.2
Maximum observed rating	5.0
Rating observations	300

KPI DEFINE USED:

Average CSAT: Mean customer satisfaction rating across the observations.

Minimum Rating: Lowest observed customer satisfaction rating.

Maximum Rating: Highest observed customer satisfaction rating.

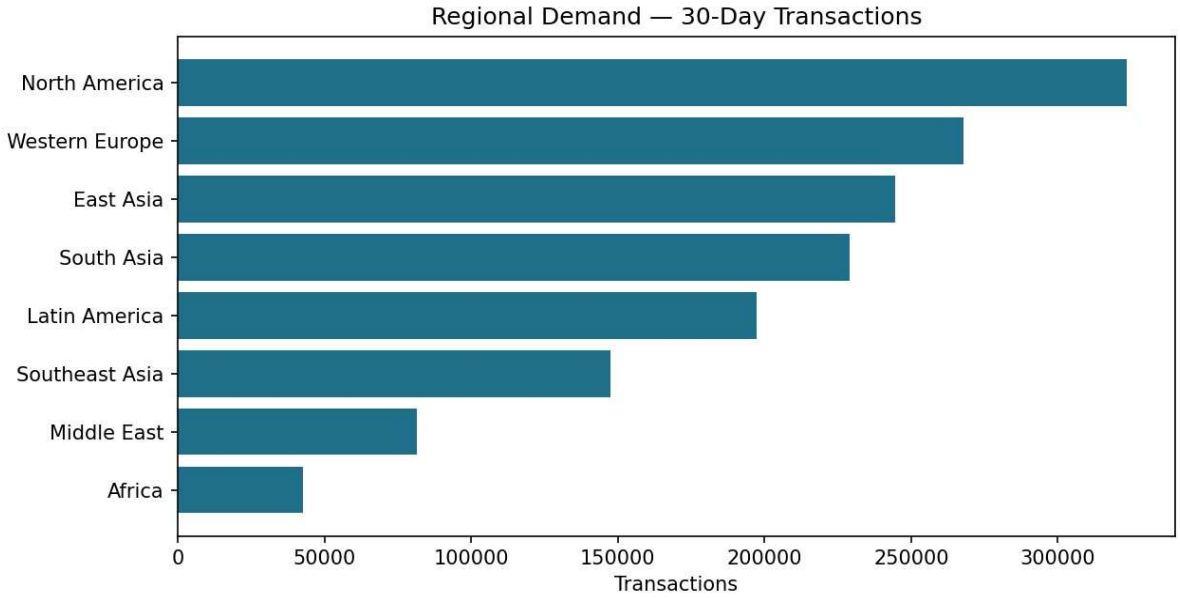
Rating Observations: Total number of CSAT observations included in the analysis.

Rating Distribution: Shows how frequently different satisfaction-rating ranges occur.

Detailed interpretation: The average rating of 4.1/5 provides an overall summary, while the histogram shows the distribution behind that average. The observed ratings range from 3.2 to 5.0. More observations in a rating range produce a taller bar. This helps show whether satisfaction is concentrated or widely spread across products and days. The graph describes rating patterns and does not establish that ratings cause higher or lower transaction volume.

7. Regional Demand

The Regional Demand graph compares 30-day transaction volume across the **8 regions** in the workbook. **North America** recorded the highest demand with 323,671 transactions (21.1%), while **Africa** recorded the lowest with 42,760 transactions (2.79%). The difference of 280,911 transactions shows that demand is not evenly distributed across regions.



Key numerical summary

Region	30-day units	Share of total
North America	323,671	21.1%

Africa	42,760	2.79%
All regions	1,534,045	100.00%

KPI DEFINE USED :

30-Day Regional Units: Total transactions recorded in each region during the 30-day period.

Regional Share (%): Percentage contribution of a region to the total transaction volume.

Highest-Demand Region: Region with the maximum transaction volume.

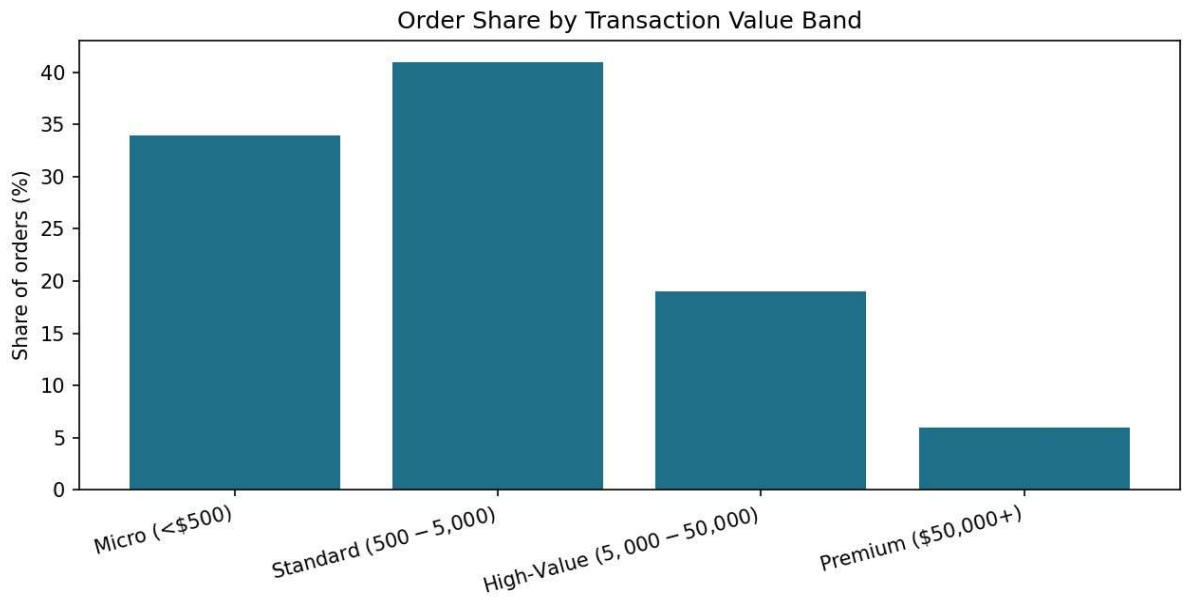
Lowest-Demand Region: Region with the minimum transaction volume.

Regional Demand Difference: Difference between the highest- and lowest-demand regions.

Detailed interpretation: North America records the highest regional demand with 323,671 transactions, equal to 21.1% of the total. Africa has the lowest with 42,760, or 2.79%. The difference is 280,911 transactions. The graph shows that demand is not evenly distributed across regions and can support descriptive regional planning and branch-capacity discussions.

8. Transaction Value Band & Order Share

The workbook contains order share by transaction value band, but it does not contain transaction-level promotional or rate-discount event data. Therefore, a discount-correlation chart or a claim that a particular rate promotion caused higher volume should not be included. The **Standard ($500-$5,000)** band has the highest order share at 41.0%, while the **Premium ($50,000+)** band has the lowest at 6.0%. This indicates that everyday, moderate-value transactions represent the largest share of activity in the dataset.



Key numerical summary

Value band	Order share
Micro (<$500)	34%
Standard ($500-$5,000)	41%
High-Value ($5,000-$50,000)	19%
Premium ($50,000+)	6%

KPI DEFINE USED :

Transaction Value Band: Classification of transactions according to their transaction-value range.

Order Share (%): Percentage of total orders belonging to each transaction-value band.

Highest Order-Share Band: Value band contributing the largest percentage of orders.

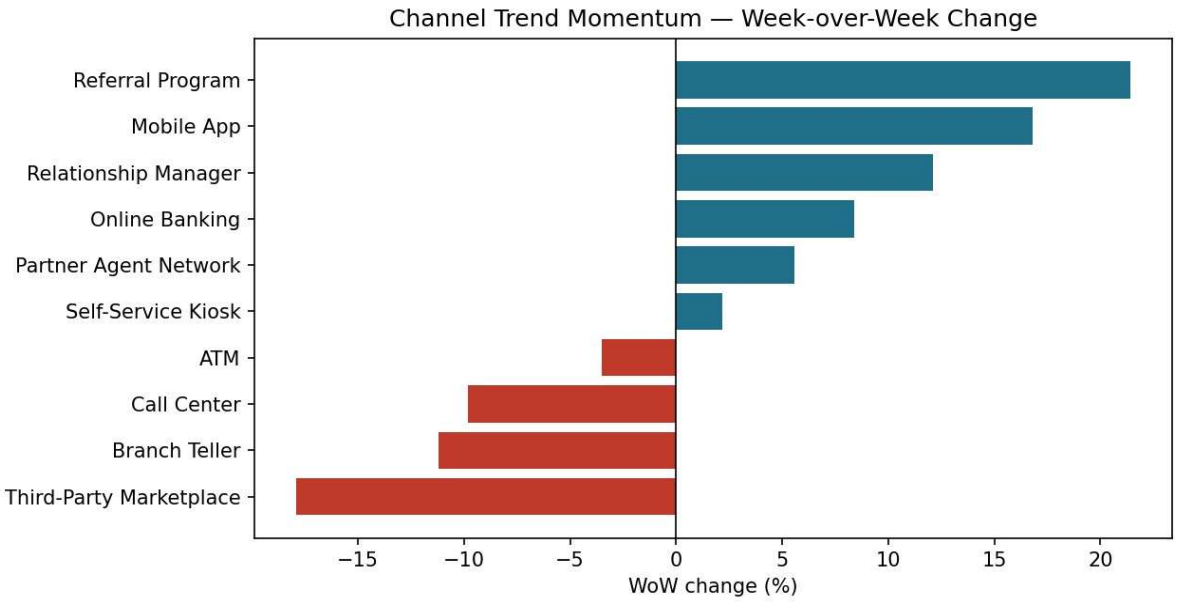
Lowest Order-Share Band: Value band contributing the smallest percentage of orders.

Share Difference: Difference in percentage points between the highest- and lowest – share bands.

Detailed interpretation: The largest share of orders is in Standard ($500-$5,000), representing 41.0% of orders. The smallest share is Premium ($50,000+), representing 6.0%. The difference is 35.0 percentage points. This is an order-share by value-band analysis, not a rate-promotion-correlation analysis, because the workbook does not provide promotion-event observations.

9. Channel Trends

The supplied Channel Trends sheet contains order share and week-over-week momentum for ten service channels. The graph therefore focuses on the channel-level trend movement directly supported by the workbook. The channel analysis shows that **Referral Program** recorded the strongest week-over-week increase of 21.4%, while **Third-Party Marketplace** recorded the largest decline of 17.9%. The highest order-share channel, **Mobile App**, represented 24.5% of orders. This analysis helps identify channels gaining or losing customer preference during the observed period.



Key numerical summary

Channel	Order share	WoW change
Referral Program	-	+21.4%
Third-Party Marketplace	-	-17.9%

Mobile App	24.5%	-

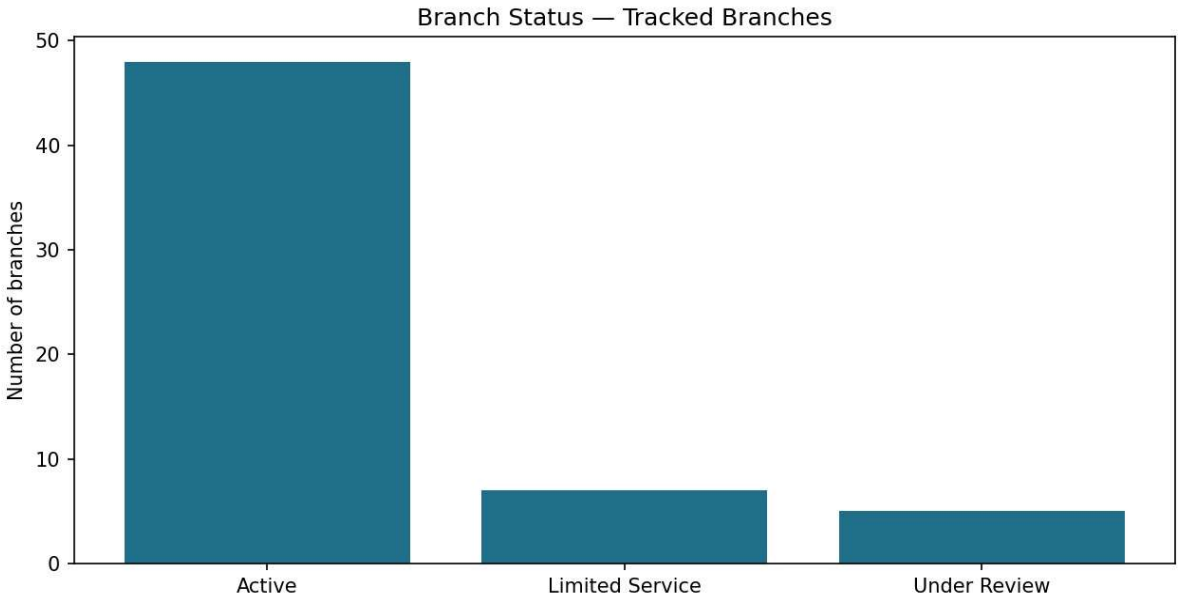
KPI DEFINE USED:

- **Order Share (%): Percentage of total orders generated through each service channel.**
- **Week-over-Week Momentum: Percentage increase or decrease in channel activity compared with the previous week.**
- **Highest-Growth Channel: Channel showing the largest positive week-over-week change.**
- **Largest-Decline Channel: Channel showing the largest negative week-over-week change.**
- **Channel Share: Relative contribution of each channel to overall orders.** □

Detailed interpretation: Referral Program has the strongest week-over-week increase at +21.4%, while Third-Party Marketplace has the largest decline at -17.9%. The difference is 39.3 percentage points. Order-share values also show the relative importance of each channel. This is a channel-level trend signal and does not prove that channel alone caused transaction changes.

10. Branch Status

The Branch Status graph summarizes the availability status of the **60 tracked branches**. The workbook provides branch status and remaining daily capacity, so the supported chart is a status-count comparison rather than a queue-time calculation. The branch status analysis shows that 48 branches (80.0%) are Active, while 7 branches (11.7%) are Limited Service and 5 branches (8.3%) are Under Review. Overall, 12 branches (20.0%) are classified as At-Risk Capacity and may require attention for operational planning.



Key numerical summary

Branch status	Branch count	Share of tracked branches
Active	48	80.0%
Limited Service	7	11.7%
Under Review	5	8.3%
Limited + Under Review	12	20.0%

KPI DEFINE USED:

Active Branches: Number of branches currently classified as Active.

Limited Service Branches: Number of branches operating under Limited Service status.

Under Review Branches: Number of branches currently marked for review.

At-Risk Capacity: Combined number of Limited Service and Under Review branches.

Branch Status Share (%): Percentage of tracked branches represented by each status category. □

Detailed interpretation: There are 48 Active branches, 7 Limited Service branches, and 5 Under Review branches. Together, 12 branches, or 20.0% of tracked branches, are either Limited Service or Under Review. This gives a clear numerical picture of branch-network health. The graph becomes more useful when read alongside regional demand and category trend graphs. Because the workbook does not provide a per-branch daily queue-time series, exact wait-time impact should not be claimed from this sheet alone.

11. Summary of Key Metrics

Metric	Value (simulated)
Total transactions (30 days)	1,538,130
Total revenue (30 days)	$29,780,780.91
Average customer rating (CSAT)	4.1 / 5
Top single-day volume	89,207 transactions (31 Jul 2026)
Fastest-rising category	Fixed Deposits (+3.7%)
Fastest-declining category	Home Mortgages (-25.0%)
Largest order-share value band	Standard ($500-$5,000) (41.0%)
Regions tracked	8
Limited Service + Under Review branches	12 of 60

Summary of Key Metrics – Explanation

The Summary of Key Metrics provides an overall view of the banking transaction performance during the 30-day reporting period. The total number of transactions was **1,538,130,** while the total fee and interest revenue generated was **$29,780,780.91**. The average customer satisfaction (CSAT) rating was 4.1 out of 5, showing a generally positive customer experience. The highest single-day transaction volume was 89,207 transactions on 31 July 2026. Among product categories, Fixed Deposits showed the fastest growth of 3.7%, while Home Mortgages recorded the largest decline of 25.0%. The largest transaction value band was Standard ($500–$5,000), accounting for 41% of orders. The analysis covered 8 regions and 60 branches, of which 12 branches (20%) were either Limited Service or Under Review. Overall, these metrics provide a quick summary of transaction volume, revenue, customer satisfaction, product trends, regional coverage, and branch status.

12. Next Steps to Move From Descriptive → Predictive

This report answers what happened and what is currently trending. The natural next layer, if useful:

- The current report focuses on descriptive analytics, which explains what has happened in the banking transaction data and identifies existing patterns and trends. The next step is to move from descriptive analytics to predictive analytics, where historical data can be used to estimate future trends and support better decision-making.

- **One important area is demand forecasting.** Historical transaction data can be analyzed to predict future demand for different products and regions. This can help banks plan staffing, branch capacity, and liquidity requirements before periods of high demand such as salary dates, festivals, or seasonal peaks.

- **Another area is product lifecycle analysis.** Predictive models can identify whether a product category is growing, approaching its peak, or starting to decline. For example, the report shows that Fixed Deposits increased by 3.7%, while Home Mortgages declined by 25%. In a predictive system, such historical movements could be used to estimate future product demand and support better product planning.

- **Anomaly detection can also be introduced.** The system could automatically identify unusual changes in transaction volume, customer satisfaction, or other key metrics. This would help the organization detect unusual patterns earlier instead of depending only on manual dashboard monitoring.

- **Another useful step is customer cohort and repeat-usage analysis.** Customers can be grouped according to when they started using a product or service and how frequently they transact. This can help distinguish genuinely new demand from existing customers who have simply increased their transaction activity.

- Overall, moving from descriptive to predictive analytics would allow the organization to go beyond understanding what happened. It would help estimate what may happen next and support more proactive planning, resource allocation, product management, and operational decision-making.